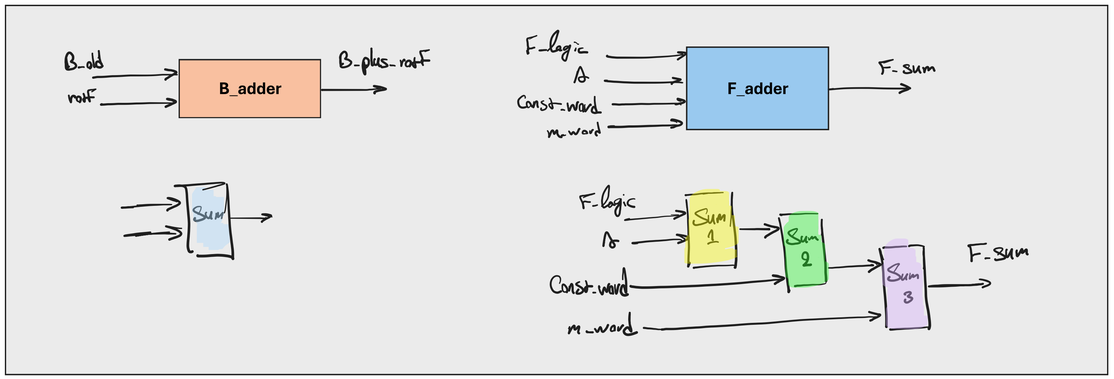
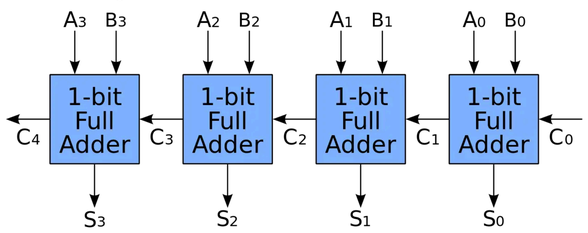
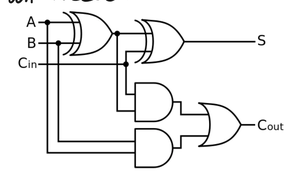
Adder:



Ripple Carry Adder:

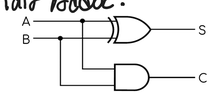


Full Adder:

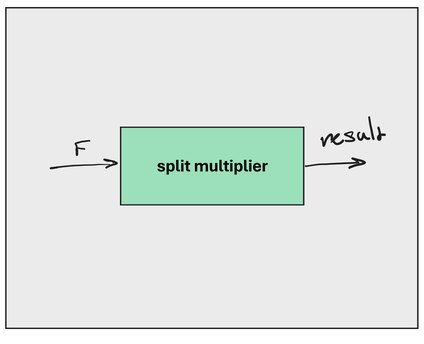


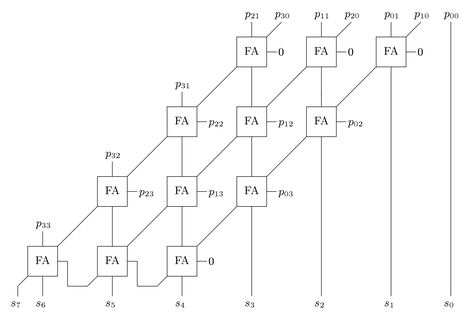
+ اونایی که C_0 ندارن به half adder می‌دارم تا گیت کمتری صرف بشه +

Half Adder:

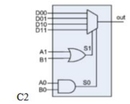


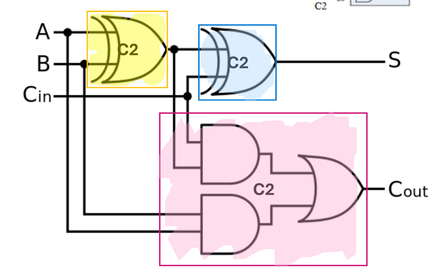
Multiplier:

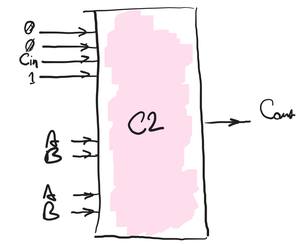




Full Adder using C2:



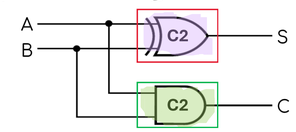




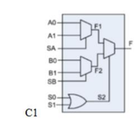
→ جایگزینی کنیم با ⇒ $C_{out} = ((A \& B) \to 1)$ else $((A \vee B) \to C_{in})$

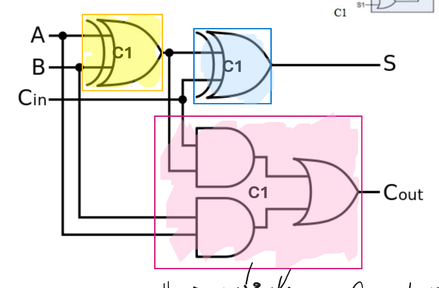
generate propagate

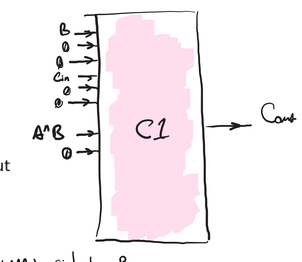
Half Adder using C2:



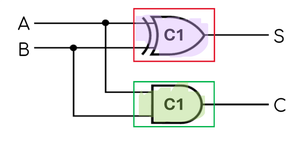
Full Adder using C1:







→ جایگزینی کنیم با ⇒ $C_{out} = ((A \wedge B) \to C_{in})$ else B

propagate generate/kill

Half Adder using C1: